HCAPO under the microscope

A faithful implementation, a length-corrected variant, and how both compare to GRPO, GiGPO and PPO across two environments

Agentic-RL-CA project

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Abstract

We evaluate HCAPO (hindsight credit assignment, arXiv:2603.08754) as published and in a length-corrected variant, on SearchQA (Qwen3-4B, full 51,713-question eval) and on ALFWorld (Qwen3-1.7B, the paper's own benchmark, unseen 134 games). Three findings. **(1) HCAPO underperforms flat GRPO in both environments** — SearchQA macro-EM 0.410–0.440 vs GRPO's 0.463; ALFWorld 0.784 vs GRPO's 0.881, where the paper reports +13.8% *over* GRPO. **(2) Every configuration we ran suffers a response-length runaway**: six SearchQA runs spanning five different interventions (ω, the ρ denominator, the injected s_{final}, and the per-turn score itself) all ignite, at steps 90/165/265/290/375/460 respectively — each knob moves *when*, none prevents *whether*. **(3) An offline diagnostic on 456 logged trajectories shows why**: the hindsight score is dominated by nuisance structure — a length correlation *created by* the injection (r=+0.43 where the plain score gives −0.02), answer-turn leakage (the final turn is top-credit in 73% of trajectories vs a 39% base rate), and a position preference that inverts with the policy's own verbosity. The length-corrected variant is by far the best-behaved (it holds normal lengths for 300 steps and ends at 1420 tokens rather than the 2048 cap) but still ignites, and still trails GRPO.

1 What HCAPO does, and the two versions we ran

After an episode ends, HCAPO re-reads each turn *conditioned on how the episode finished* and up-weights the turns that look pivotal in hindsight. Formally (Eq. 6–7 of the paper), each turn is scored by a per-token-mean log-probability under a prompt augmented with s_{final}, and credited by its score relative to its own trajectory's mean:

$$m_t = \frac{1}{|a_t|}\sum_j \log \pi_\theta(y_j \mid y_{<j}, s_t, s_{final})\,, \qquad \rho_t = \mathrm{clip}\left(\frac{e^{m_t/T}}{\frac{1}{T}\sum_k e^{m_k/T}}, 0.8, 1.2\right), \qquad Q_t^H = \rho_t\,\gamma^{T-1-t}R.$$

Q^H is group-normalised over the prompt's rollouts and added, with weight ω, to the standard GRPO term. We ran two scoring rules:

- **"hind"** — m_t exactly as above (the published form).
- **"lift"** — m_t replaced by the *two-pass difference* $\Delta m_t = \frac{1}{|a_t|}\sum_j[\log \pi(y_j|\cdot, s_{final}) - \log \pi(y_j|\cdot)]$, everything else unchanged. Motivation: subtract the shared per-token predictability profile so only the hindsight information remains.

crossed with two choices of s_{final} (the trajectory's last retrieved-documents block, or the agent's own final answer), giving a 2×2. A prior implementation of ours divided by the on-policy probability instead of the trajectory mean; that was a genuine bug, fixed before these runs, and its two runs are retained here only as additional points on the ignition curve.

2 Issue 1: an unconditional response-length runaway

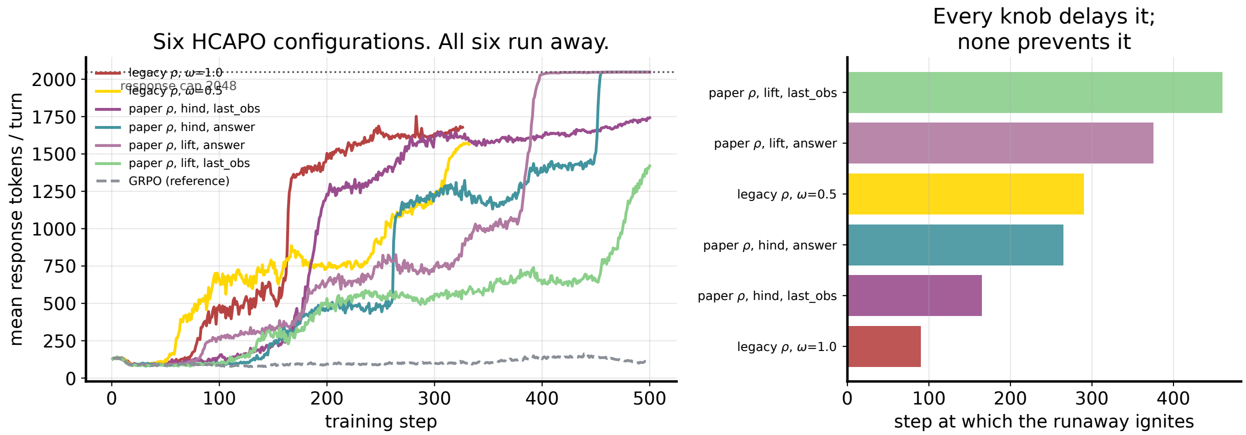

Figure 1: *Left:* mean response length per training step for six HCAPO configurations, with GRPO dashed for scale. Each behaves normally for a while, then inflates 3–10× within ∼30 steps and pins against the 2048-token cap. *Right:* the ignition step, ordered. Five distinct interventions — halving ω, fixing the ρ denominator, shrinking the injected s_{final}, and changing the per-turn score — each delay ignition; none prevents it.

Mechanism. m_t is an *average* over tokens. A turn's informative tokens (the search query, the decision) are the low-probability ones; continuation filler is high-probability. Padding therefore raises a turn's average score without adding content. Eq. 7's trajectory-mean normalisation cancels *uniform* lengthening — we verified this is exactly true — but training never moves uniformly: in each batch the turn that happens to be longer than its siblings scores $\rho > 1$ and is reinforced. The comparison re-centres every step, so no trajectory ever nets credit, yet the policy shift is permanent. Grading on a curve does not stop grade inflation. The race ends only when every turn is maximally padded, at which point $\rho \to 1$ everywhere and the hindsight term is inert: we measure the spread σ_ρ rising while the ratchet is actively selecting ($0.062 \to 0.094$) and then collapsing after ignition ($\to 0.052$; in the last_obs run $0.135 \to 0.030$).
Cost. At the endpoint, 65–100% of turns are truncated mid-sentence, training steps take ∼2.6× longer, and — because a truncated turn emits no action — 36–53% of *evaluation* episodes exhaust the 4-turn budget without answering.

3 Issue 2: the credit signal tracks nuisance, not pivotal-ness

We re-scored 456 logged trajectories offline (no training) under matched conditions: with the hindsight text, without it, and with the *gold answer* injected instead, in two regimes — the pre-RL policy (healthy lengths) and HCAPO at step 250 (post-runaway).
Two further results from the same diagnostic: Eq. 7's ranking and the lift's ranking agree at Spearman 0.95–0.97 (top-turn agreement 94–97%), and replacing the retrieved documents with the literal gold answer preserves ∼72% of the ranking. A signal whose ordering barely changes when you change *what the hindsight says* is not primarily carrying hindsight information.

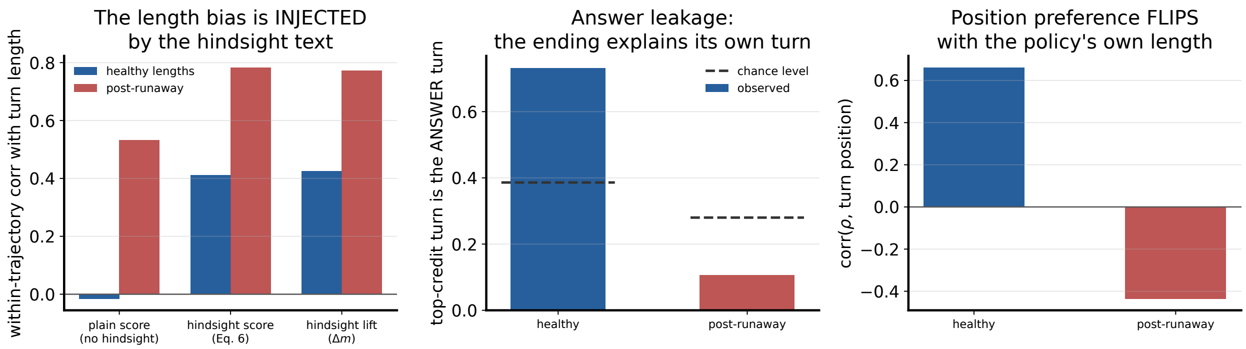

Figure 2: *Left:* within-trajectory correlation between a turn's score and its length. The plain score shows none (−0.02); adding the hindsight text creates +0.41, and the lift — which should isolate hindsight information — carries *all* of it (+0.43, surviving position controls). *Middle:* how often the top-credit turn is the final (answer) turn — 73% in the healthy regime against a 39% base rate, because the ending trivially entails the turn that produced it. This survives both the lift and gold-answer injection. *Right:* the preference for early vs late turns *inverts* between regimes (+0.66 → −0.44).

Table 1: SearchQA, full 51,713-question eval, Qwen3-4B. Non-thinking protocol except where starred. "tok/q" = generated tokens per question.

method	macro-EM	tok/q	mean turns
GRPO	**0.463**	444	3.99
GiGPO*	0.446	1680	3.58
HCAPO (hind, answer)	0.440	7719	3.77
HCAPO (lift, last_obs)	0.429	5107	3.54
turn-PPO	0.417	556	2.61
HCAPO (hind, last_obs)	0.410	6801	3.91
token-PPO*	0.381	2502	2.78
pre-RL floor	0.313	1317	1.77

4 Performance against GRPO, GiGPO and PPO

ALFWorld (Qwen3-1.7B, RL from a shared replay-BC policy, unseen 134 games): turn-PPO 0.963, GiGPO 0.955, GRPO 0.881, **HCAPO (paper-correct)** 0.784, token-PPO 0.761. The corrected implementation did improve *training*-set behaviour markedly over our buggy one (seen-val peak 0.922 vs 0.859) — but the advantage vanished on held-out games (0.784 vs 0.791), i.e. it generalised no better. Its failure is concentrated on the longest-horizon task type, `pick_two_obj_and_place`: 0.308, against 1.000 for both GiGPO and turn-PPO — precisely where per-turn credit should help most.

5 Did the length-corrected variant help?

Partly, and instructively. Against its own control (same s_{final}, only m_t differs):

score on s_{final}=last_obs	ignition step	final tokens	final clip
hind (published)	165	1742	0.75
lift (Δm)	460	1420	0.65

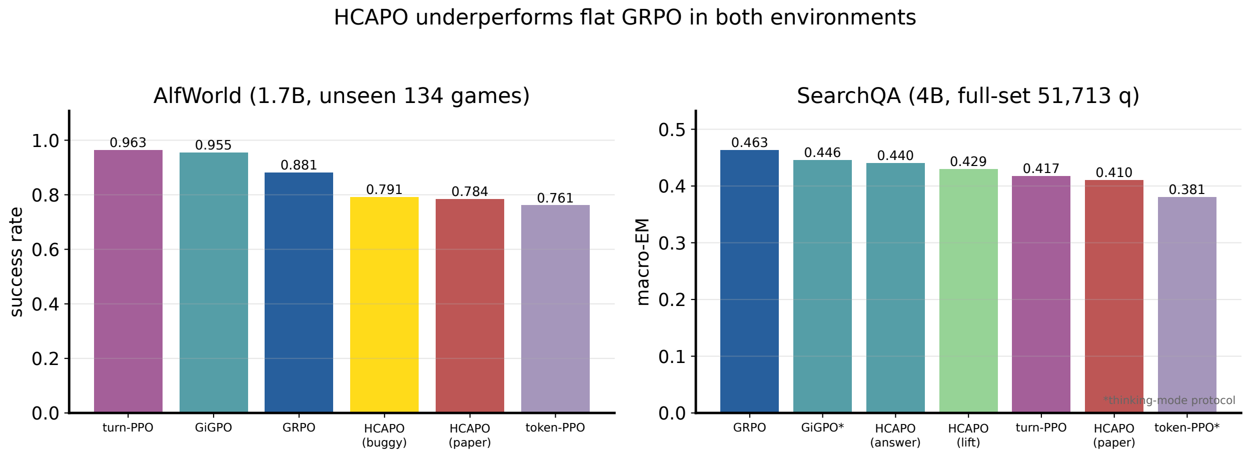

Figure 3: Both environments, identical harnesses within each. ALFWorld is the HCAPO paper's own benchmark, where it reports +13.8% over GRPO; we measure −9.7 points. On SearchQA the best HCAPO variant trails GRPO by 2.3 points.

It held ∼530–650 tokens for 300 consecutive steps — six flat measurements — and posted the best mid-training accuracy of any HCAPO run (0.447 at step 350, level with GRPO's endpoint). Then it ignited between steps 450 and 500. Full-set macro-EM 0.429: better than the published form's 0.410, worse than the answer-injection form's 0.440, and below GRPO's 0.463. The lift×answer cell completed after this report was first compiled: final val-EM 0.411 at the cap (2047 tokens, clip ≈1.0), confirming its ∼step-375 ignition; its full-set number will be appended when its eval lands.

The static diagnostic had predicted the lift would behave *identically* to Eq. 6 (Spearman 0.97 co-ranking). It did not — so rank-equivalence at a policy snapshot does not imply equivalent training dynamics. That is a genuine, reusable finding about how to evaluate credit estimators offline: snapshot rankings are insufficient; the feedback loop must be simulated.

6 Verdict and what would actually fix it

HCAPO as published did not reproduce in our hands, on its own benchmark or ours, with an implementation we verified line-by-line against Eq. 6–7 (ρ centred on 1.0, two-sided, both clip bounds engaging). Its per-turn signal is dominated by length, leakage and position artifacts, and it carries a length ratchet that no parameter-level intervention removed.

Every intervention we tested changes the *gain* of the length gradient. The untested class is the one that changes its *direction*: score turns on a **fixed window** (or another length-invariant statistic) so that padding cannot move m_t at all; and **exclude the final turn** from hindsight credit, which our measurements show is the top-credit turn 73% of the time for reasons of entailment rather than pivotal-ness.

Fine print

Single seed per configuration. SearchQA: Qwen3-4B, non-thinking, 4-turn cap, 2048-token responses, 500 steps, identical harness for every arm; the GiGPO and token-PPO rows use the thinking protocol and are marked. ALFWorld: Qwen3-1.7B, transcript prompt, 512-token cap, 150 steps, evaluated at best-seen checkpoint (the same rule for every arm). The ALFWorld number selects step 145 on a 128-game validation, which is itself noisy. The

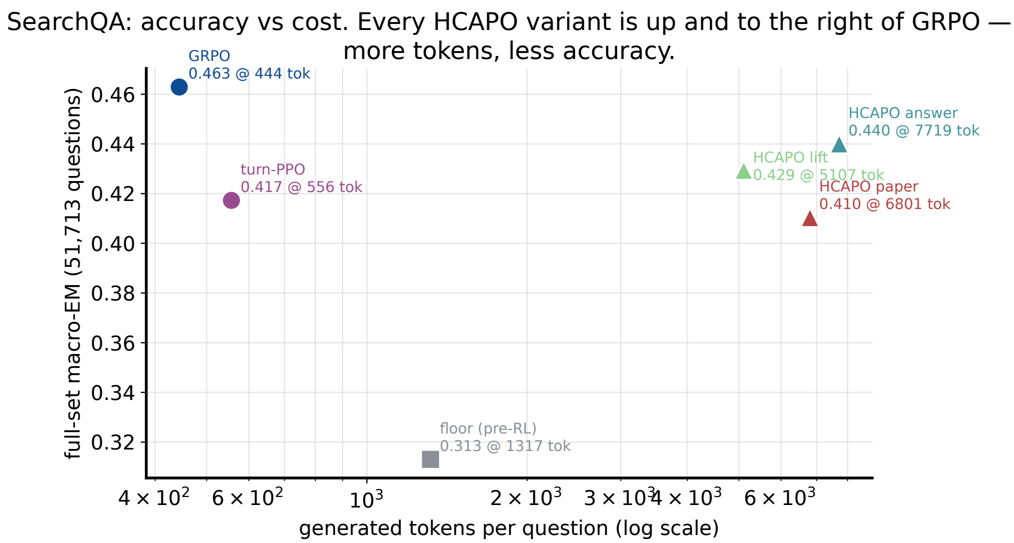

Figure 4: Accuracy against token cost. Every HCAPO variant sits up and to the right of GRPO: more tokens, less accuracy. The best HCAPO configuration spends 17× GRPO's tokens to score 0.023 lower.

HCAPO paper's benchmarks use short action-style turns under tight caps and a different base model; our results are a statement about these two task settings, not a general refutation. Data, figures and the rerunnable analysis: `assets/build_figures.py`; the complete timestamped trail — including five premature stability calls I recorded and retracted — is in `docs/reports/2026-07-21_4b_horizon_eval/results/hcapo_wave3_preregistration.md`. Companion documents: `2026-07-29_hcapo_implementation_note/` (the ρ denominator bug), `2026-07-30_hcapo_length_runaway/` (the runaway in plain language), and `2026-07-21_4b_horizon_eval/` (the GRPO/GiGPO/PPO cross-environment study).

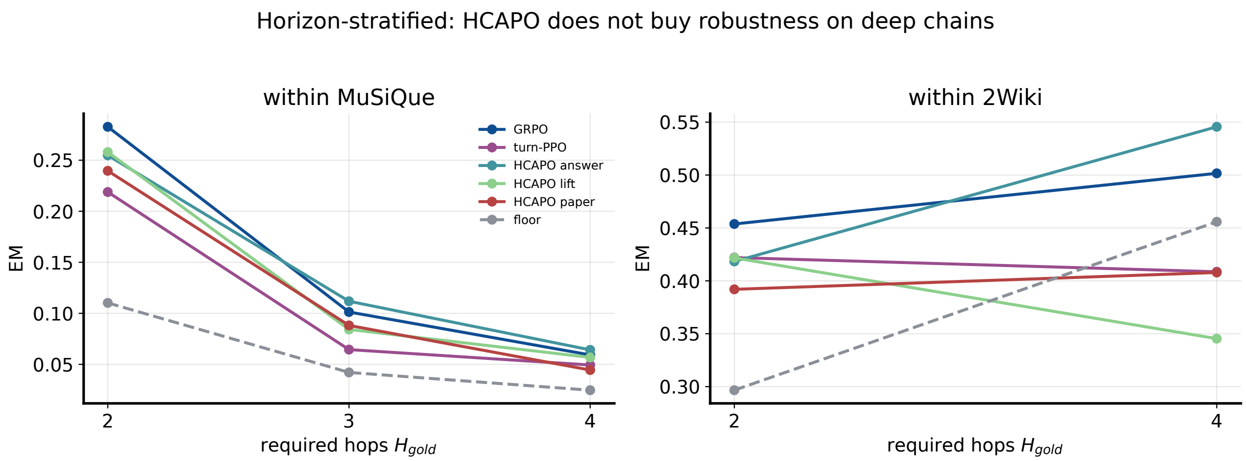

Figure 5: Horizon-stratified SearchQA EM within the two multi-hop-spanning tasks. HCAPO tracks or trails the baselines at every hop count; the deep-chain collapse (MuSiQue 4-hop) is not mitigated.